

Course: GRC103 Risk Assessment and Management Techniques

Lab: Practical Risk Monitoring with Wazuh

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1. Wazuh Agent Successfully Connected

The Wazuh Agent was installed on the Windows host machine and successfully registered with the Wazuh Manager running on the Ubuntu VirtualBox server. The dashboard confirmed the agent status as Active, demonstrating that the risk monitoring platform was fully operational and the endpoint was being monitored in real time.

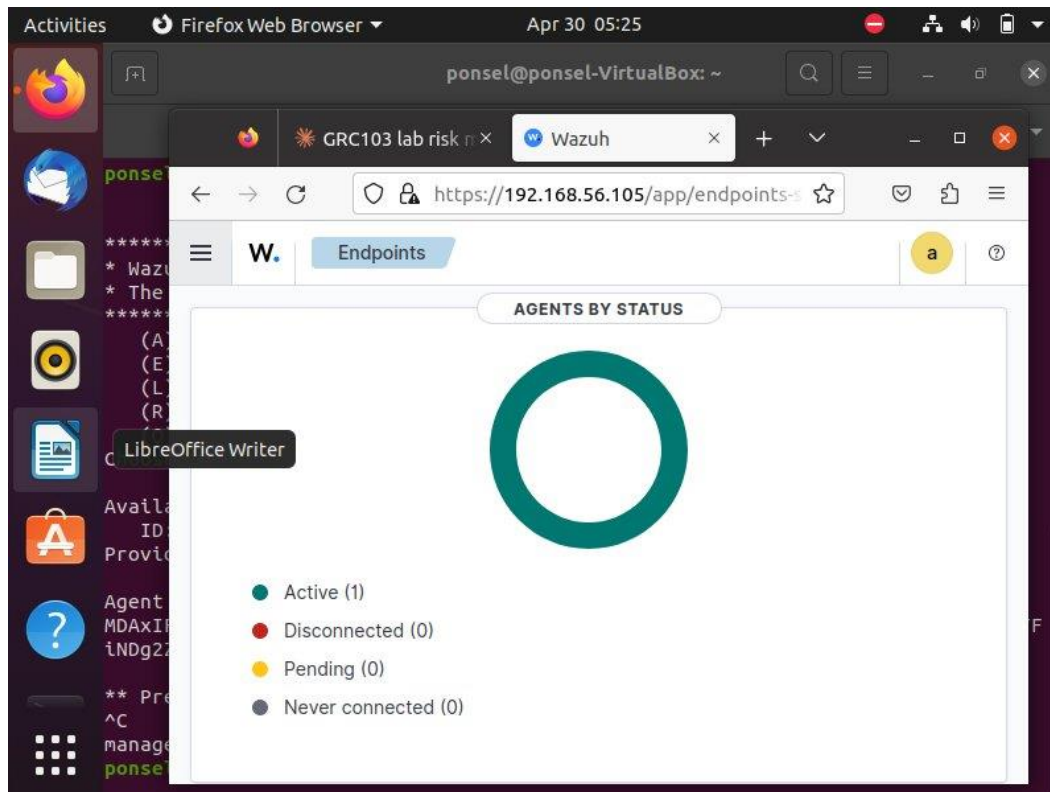


Figure 1: Wazuh Endpoints dashboard showing Windows-PC agent with Active status

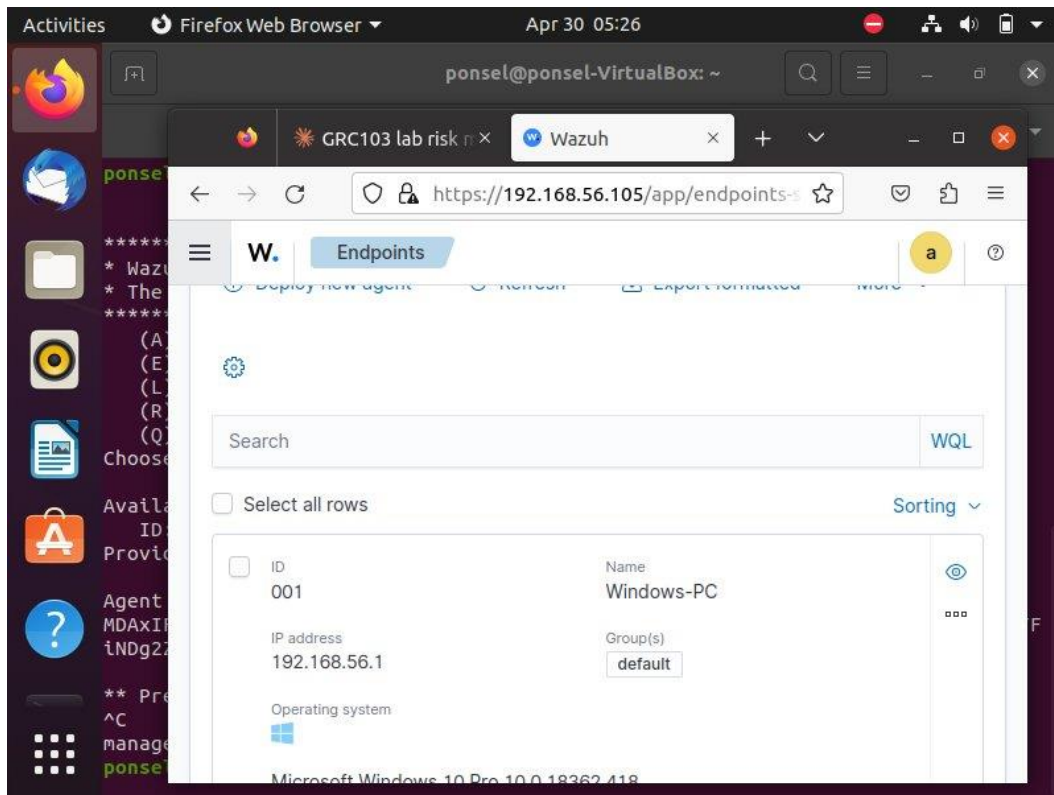


Figure 2: Agent details showing Windows-PC (001), IP address 192.168.56.1, running Microsoft Windows 10 Pro

2. File Integrity Monitoring (FIM) – Detective Control Implemented

The File Integrity Monitoring (FIM) module was configured as the detective control by editing the `ossec.conf` file on the Windows agent. The `SensitiveFiles` directory was added to the `syscheck` section with realtime monitoring enabled. This configuration instructed the Wazuh agent to continuously watch the designated folder and report any file creation, modification, or deletion to the manager.

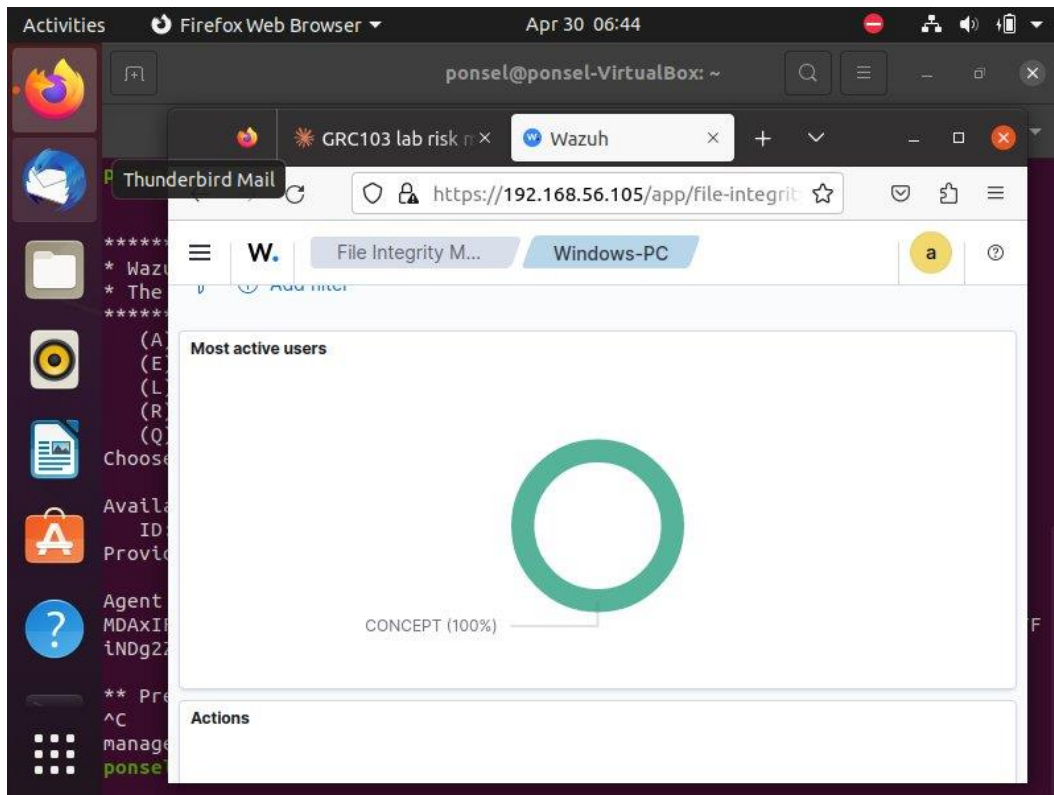


Figure 3: File Integrity Monitoring dashboard showing events – added, deleted, and modified – on the monitored SensitiveFiles folder under C:\Users\CONCEPT

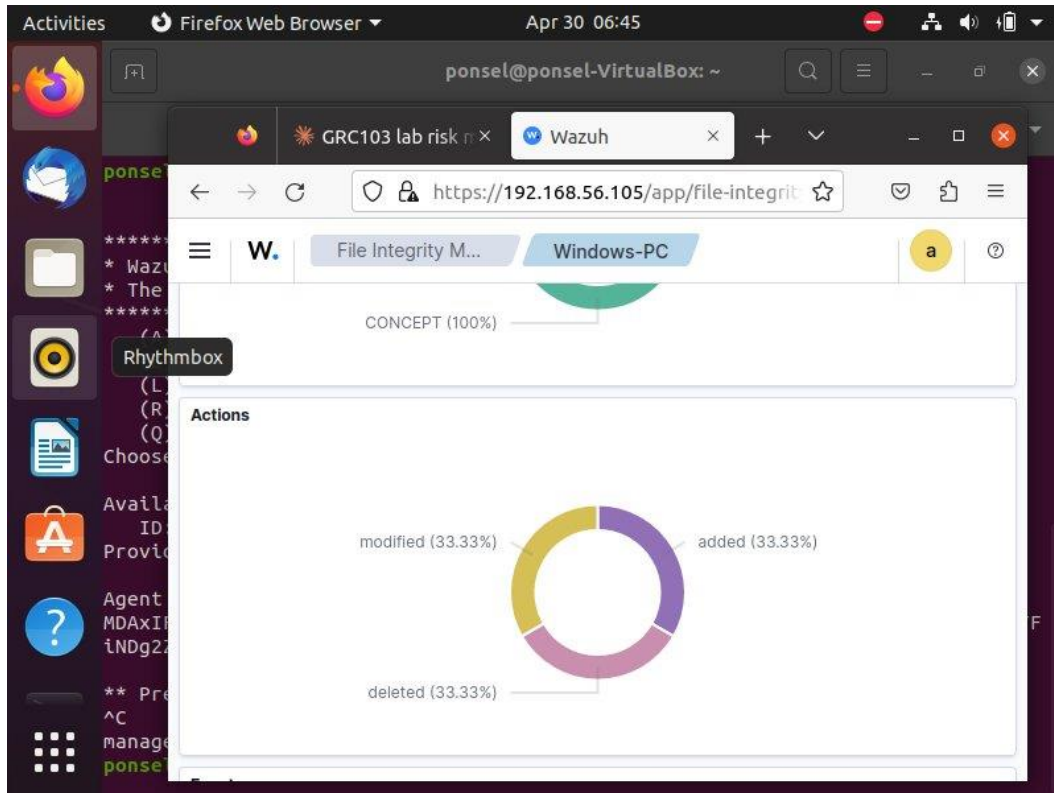


Figure 4: Actions breakdown showing equal distribution of added (33.33%), modified (33.33%), and deleted (33.33%) file events, with CONCEPT identified as the most active user

3. Risk Events Detected Alerts Generated

To verify that the detective control was operating effectively, three risk events were simulated inside the SensitiveFiles folder: a file was created, modified, and then deleted. Wazuh detected all three actions and generated a total of 6 alerts in the Threat Hunting module filtered by rule.groups: syscheck.

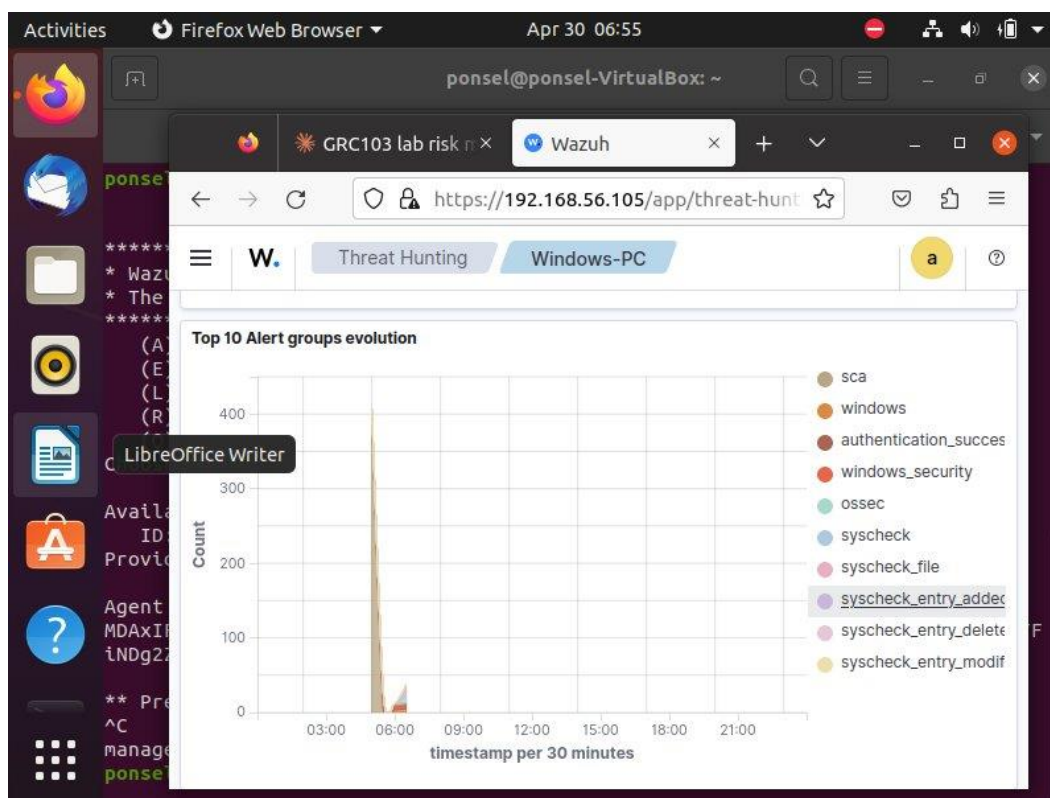


Figure 5: Wazuh Threat Hunting dashboard showing 6 total syscheck alerts for Windows-PC agent

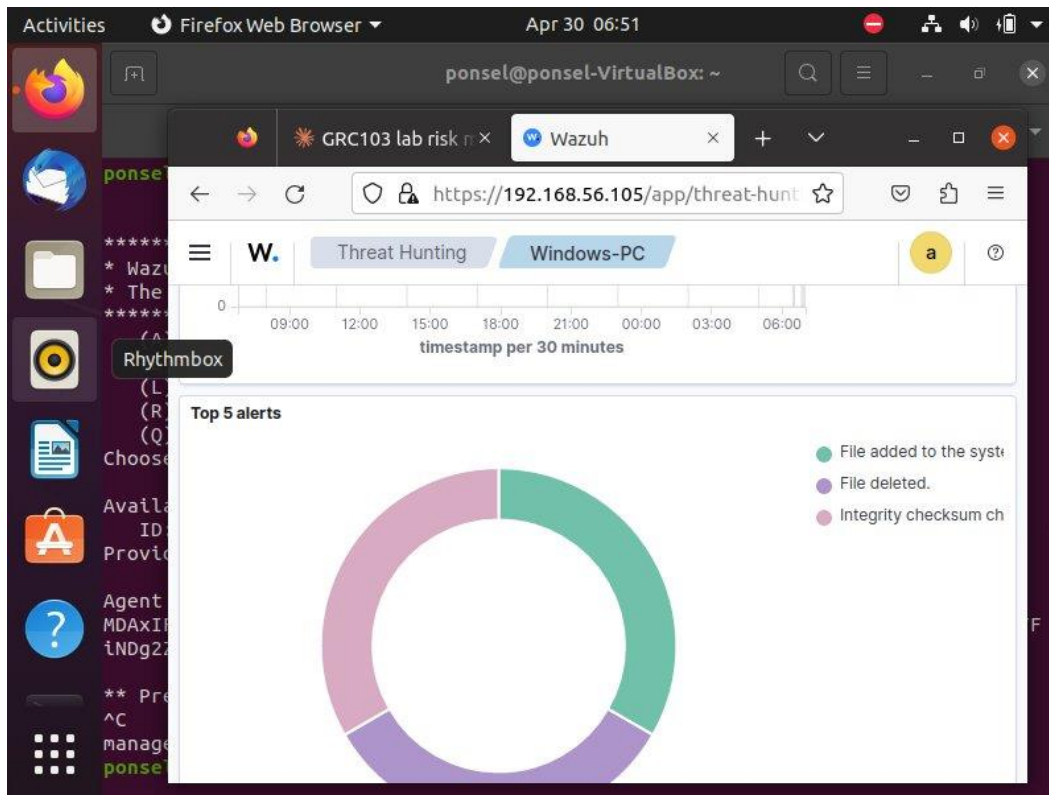


Figure 6: Alert timeline showing spike of activity at approximately 06:00, corresponding to the simulated file events

4. Alert Details Rule IDs and Timestamps

Each alert was associated with a specific rule ID that identified the type of file event. Rule 554 captured file additions, Rule 553 captured file deletions, and Rule 550 captured integrity checksum changes (modifications). The timestamps confirmed the alerts were generated in real time, providing a verifiable audit trail of all file activity.

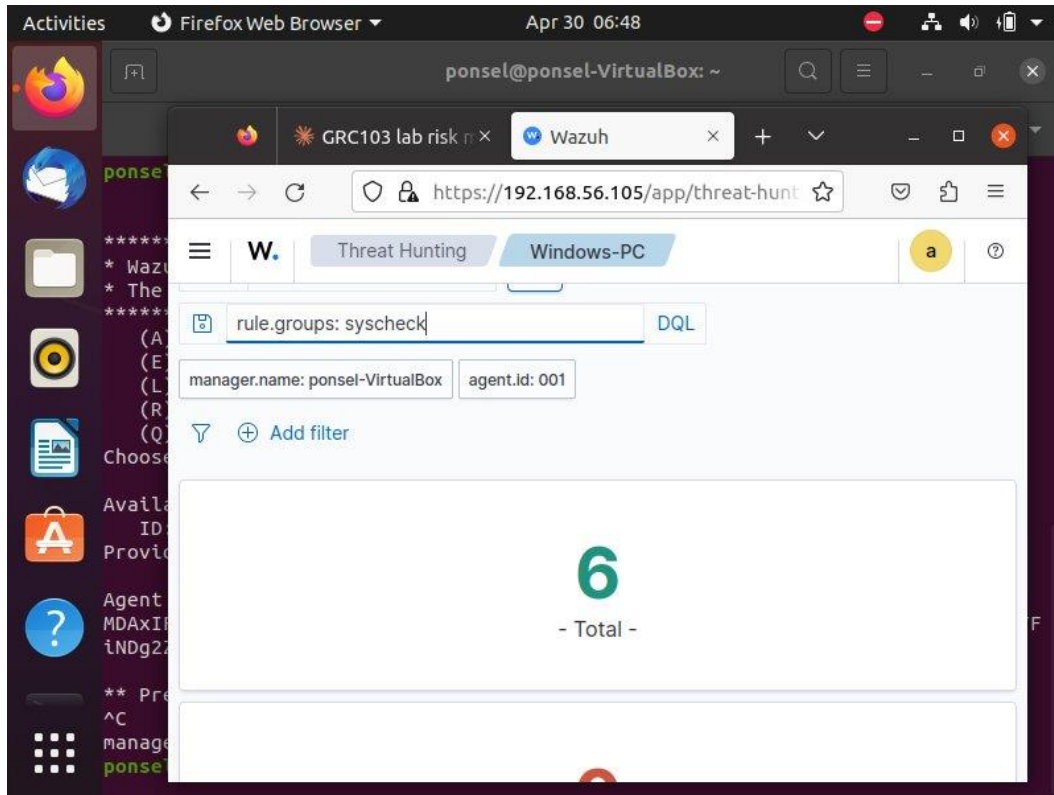


Figure 7: Alert details showing Rule IDs, descriptions, severity levels, and counts for each type of file event detected

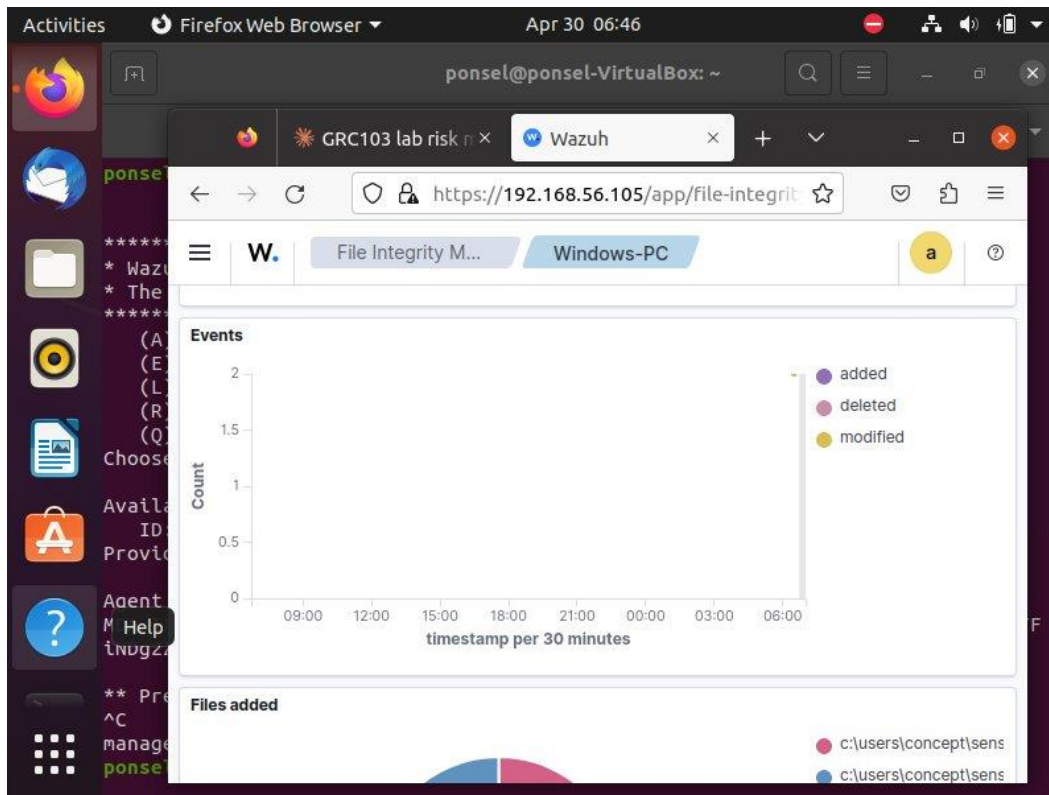


Figure 8: Timestamps on individual alerts confirming real-time detection of file events on April 30, 2026

5. Wazuh Generated Report Evidence for Compliance

A formal Threat Hunting Report was generated directly from the Wazuh dashboard. The report provides a comprehensive summary of all security alerts, rule group activity, and file integrity events. This type of report serves as documentary evidence for compliance audits under frameworks such as SOX, PCI DSS, and HIPAA, demonstrating that a detective control was in place and operating effectively during the monitored period.

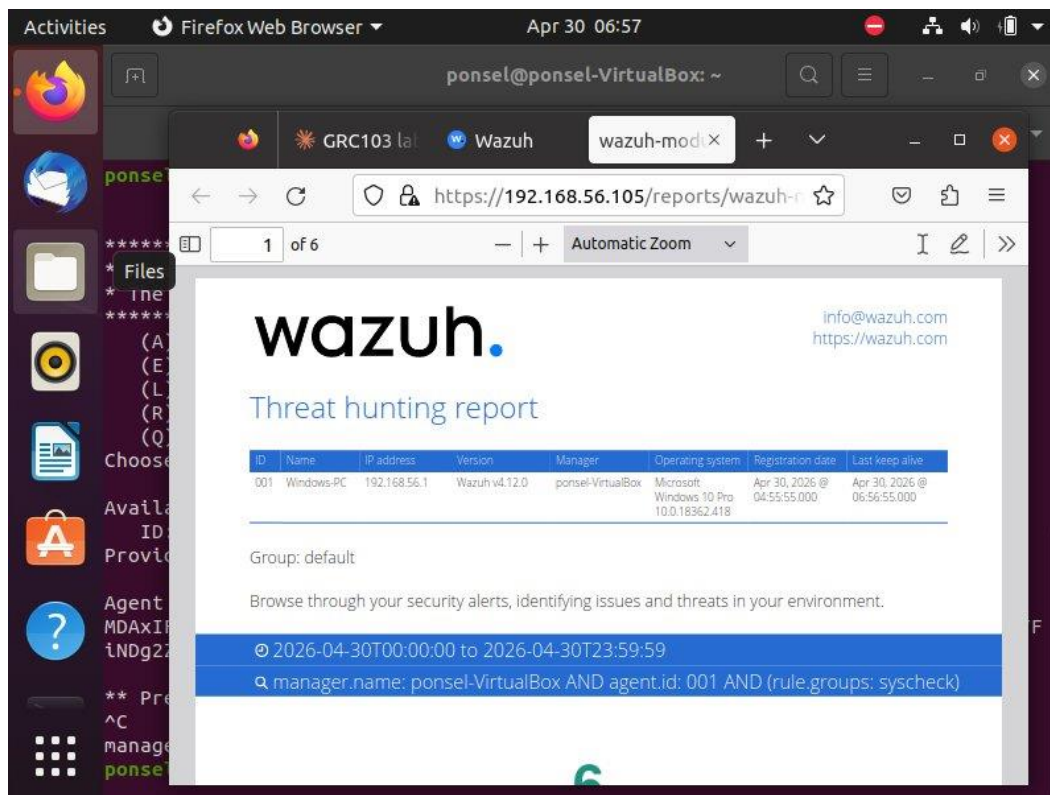


Figure 9: Wazuh Threat Hunting Report cover page showing agent details, monitoring period, and total of 6 alerts detected

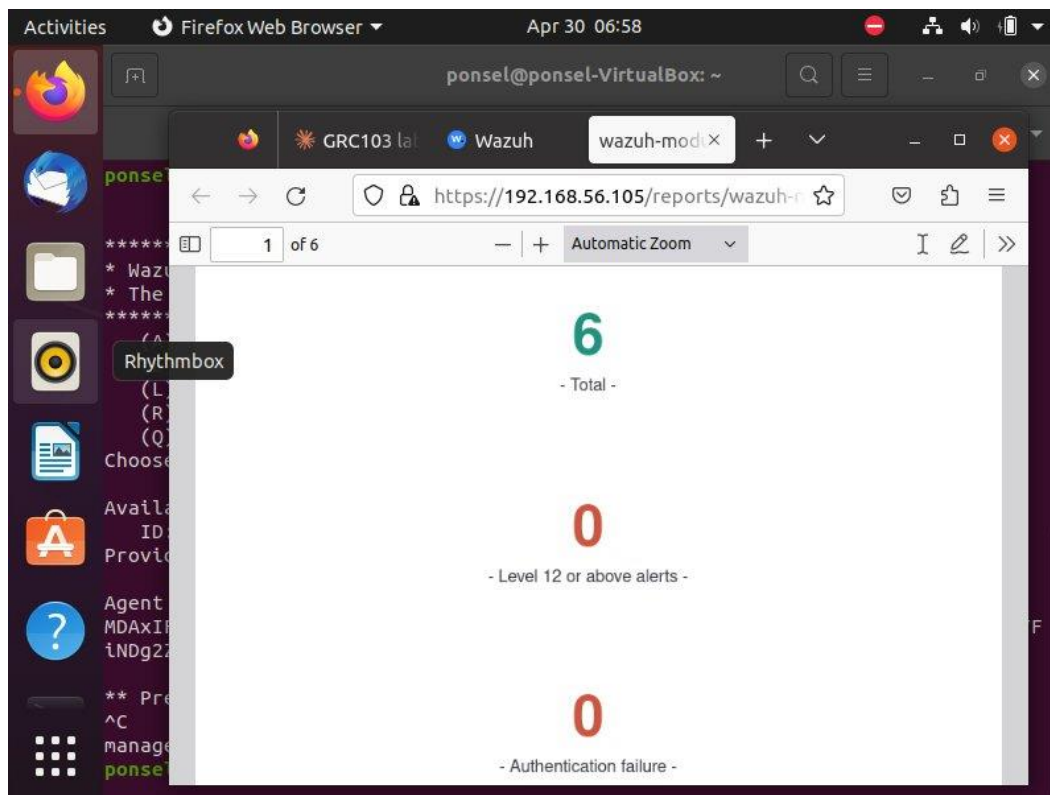


Figure 10: Report page showing Top 10 Alert Groups evolution timeline and authentication summary

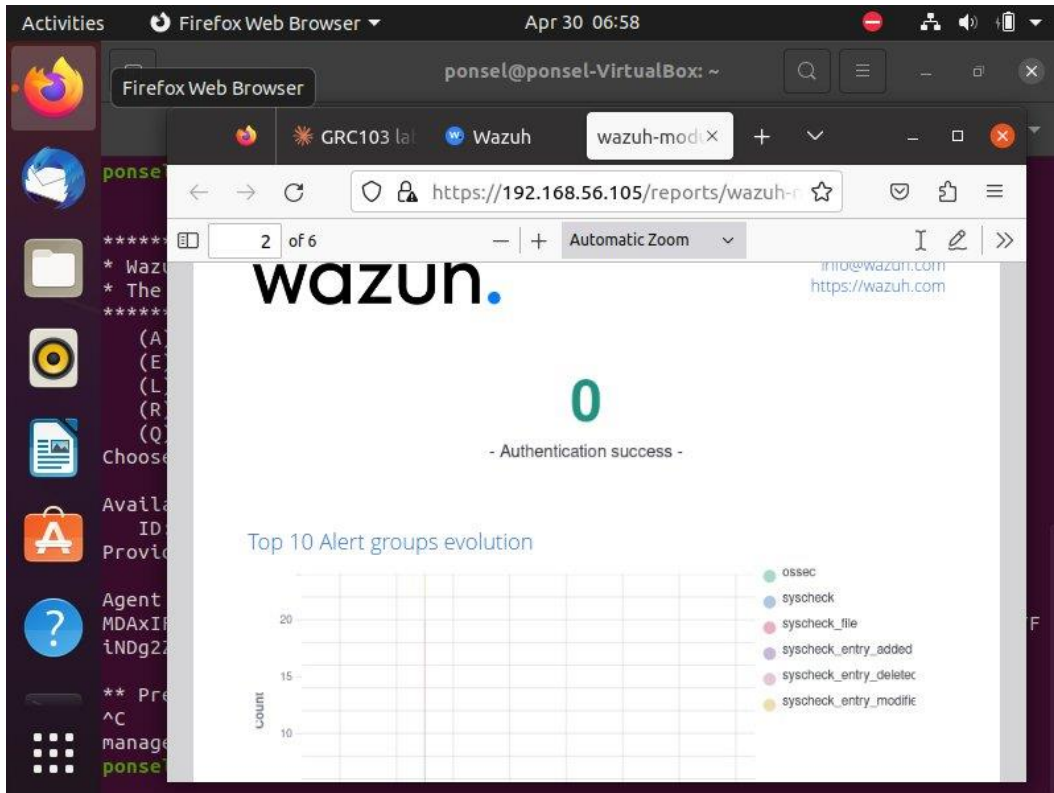


Figure 11: Report page showing Alerts timeline chart and Top 5 alerts breakdown by type

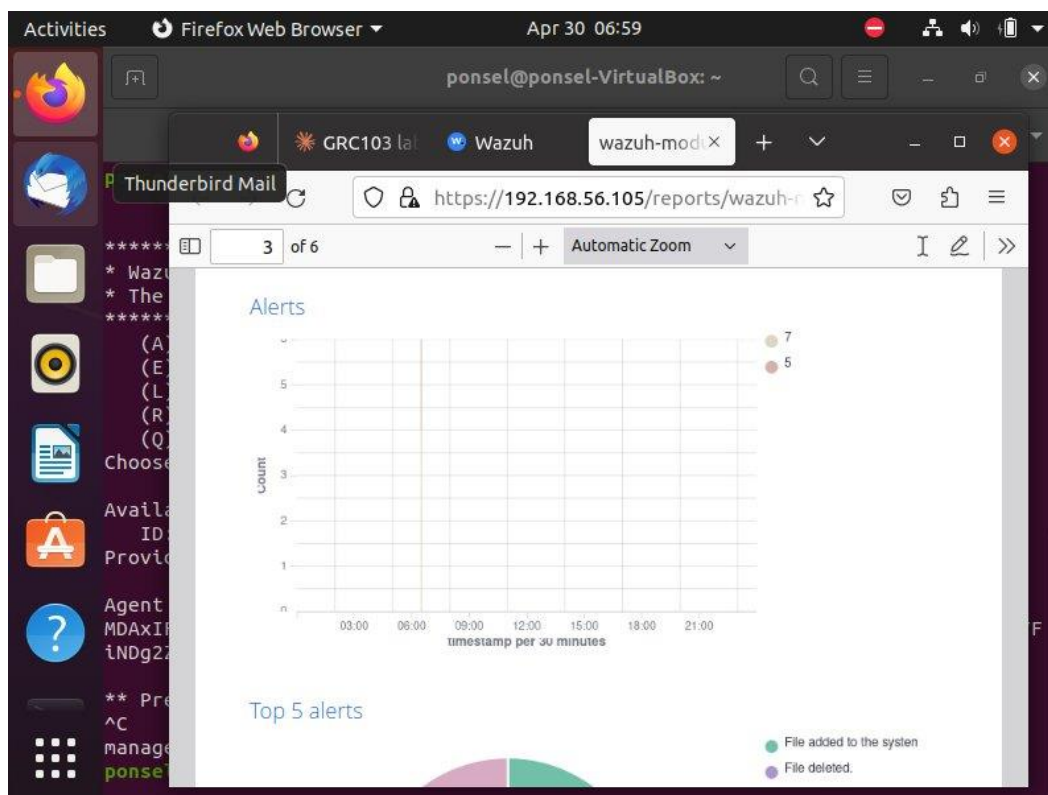


Figure 12: Alerts Summary table – Rule 550 (Integrity checksum changed, Level 7), Rule 553 (File deleted, Level 7), Rule 554 (File added, Level 5), each triggered 2 times

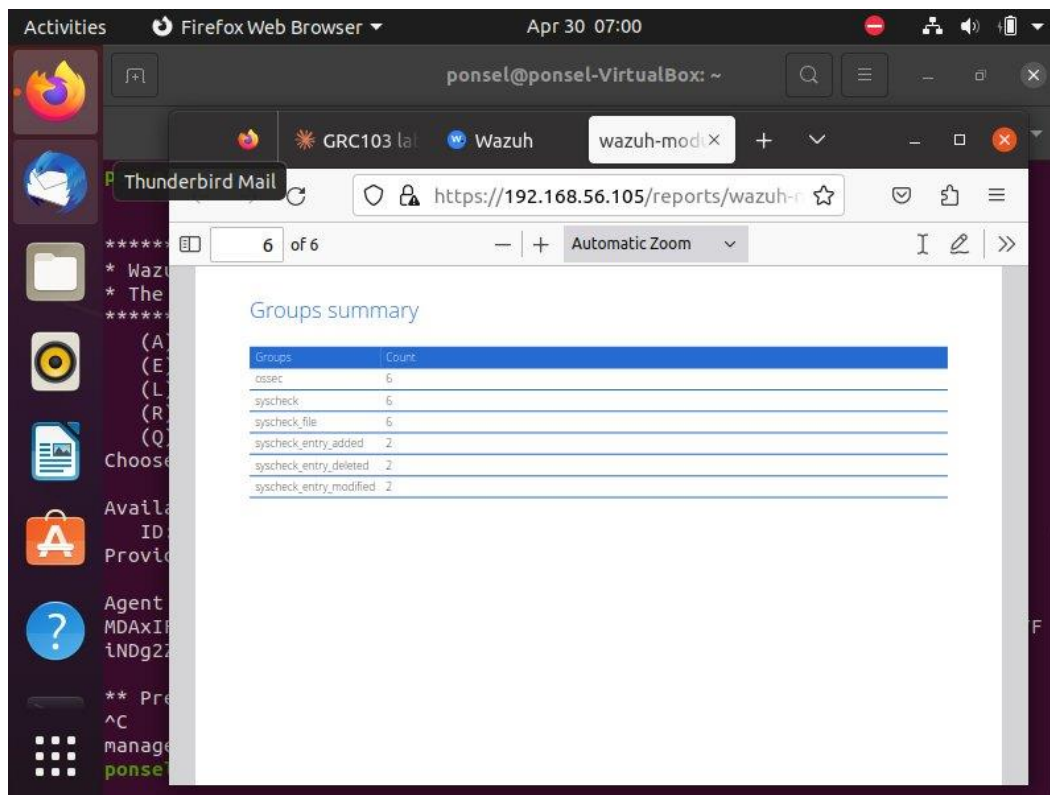


Figure 13: Groups Summary showing ossec (6), syscheck (6), syscheck_file (6), syscheck_entry_added (2), syscheck_entry_deleted (2), syscheck_entry_modified (2)

6. GRC Workshop Analysis Questions

6.1 Risk Identification and Analysis

The risk identified in this scenario is the unauthorized modification or deletion of sensitive financial forecast documents. If this risk were realized, the business impact could be severe across multiple dimensions. Financially, tampered forecast documents could lead to wrong investment decisions, misreported earnings, or regulatory fines. Reputationally, if stakeholders or auditors discovered that financial data had been altered without authorization, trust in the organization would be damaged significantly. Operationally, staff would be unable to rely on the integrity of documents used for planning and decision making, causing delays and errors in business processes.

As seen in the Wazuh dashboard and the generated report, the control helped quantify the risk by capturing exactly 6 alerts in total, with the actions broken down equally into added (33.33%), modified (33.33%), and deleted (33.33%). This means the system detected and

logged every single file event in real time, giving management a measurable frequency of activity on sensitive files.

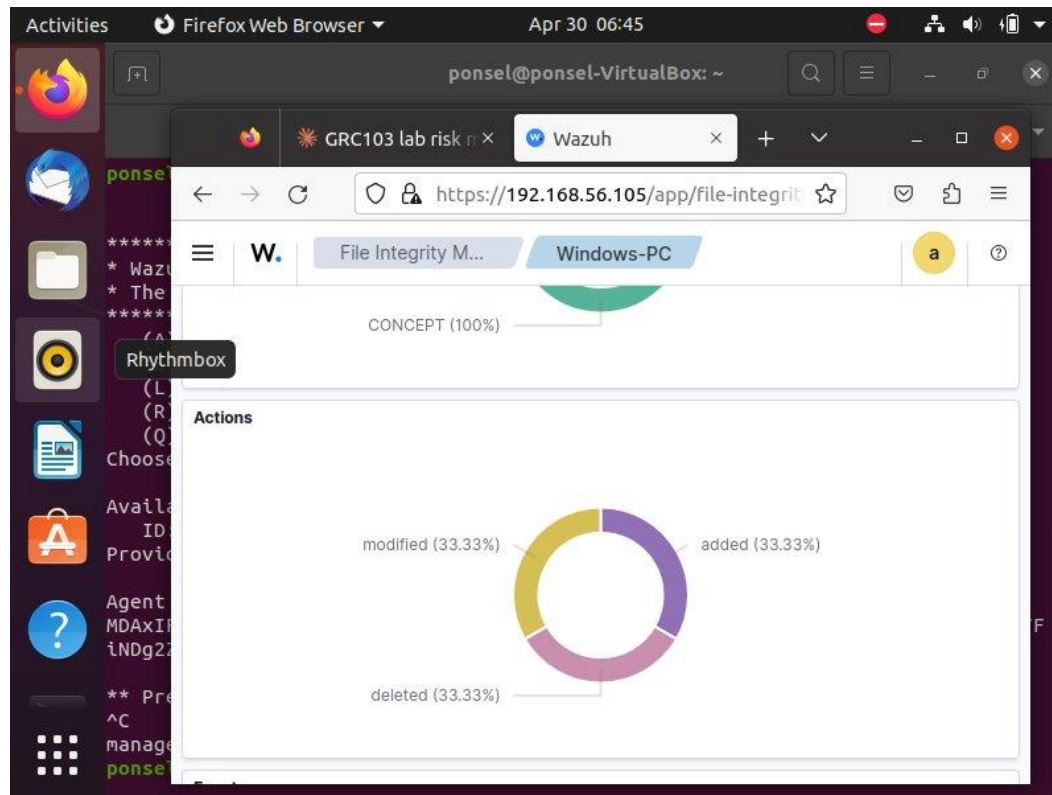


Figure 14: Actions pie chart showing equal distribution of file events – evidence of risk quantification

6.2 Control Evaluation and Treatment

A complementary preventive control for this same risk would be implementing strict file permission settings on the SensitiveFiles folder, allowing only authorized users to write or delete files. This would stop unauthorized changes before they happen, whereas the Wazuh FIM control detects them after they occur. For example, if the Wazuh dashboard generated an alert showing that a financial document was deleted outside of business hours by an unexpected user account, this alert would immediately trigger a corrective control such as restoring the file from backup, revoking the user's access, and launching an internal investigation.

6.3 Risk Monitoring and Reporting

For reporting to a manager on the operational effectiveness of this FIM control, three key metrics visible in the Wazuh dashboard would be: the total number of file integrity alerts generated (which was 6 in our case as shown on the report), the breakdown of alert types by action such as files added, deleted, and modified as shown in the alerts summary table with Rule IDs 550, 553, and 554, and the timestamp of each event showing exactly when the activity occurred as displayed in the alerts timeline graph.

Centralizing this monitoring data on the Wazuh manager rather than checking logs on each individual computer greatly improves governance and auditability because it creates a single source of truth that cannot easily be tampered with by the same user performing the actions. An auditor or manager can log into one dashboard and see a complete, time-stamped audit trail across all monitored endpoints.

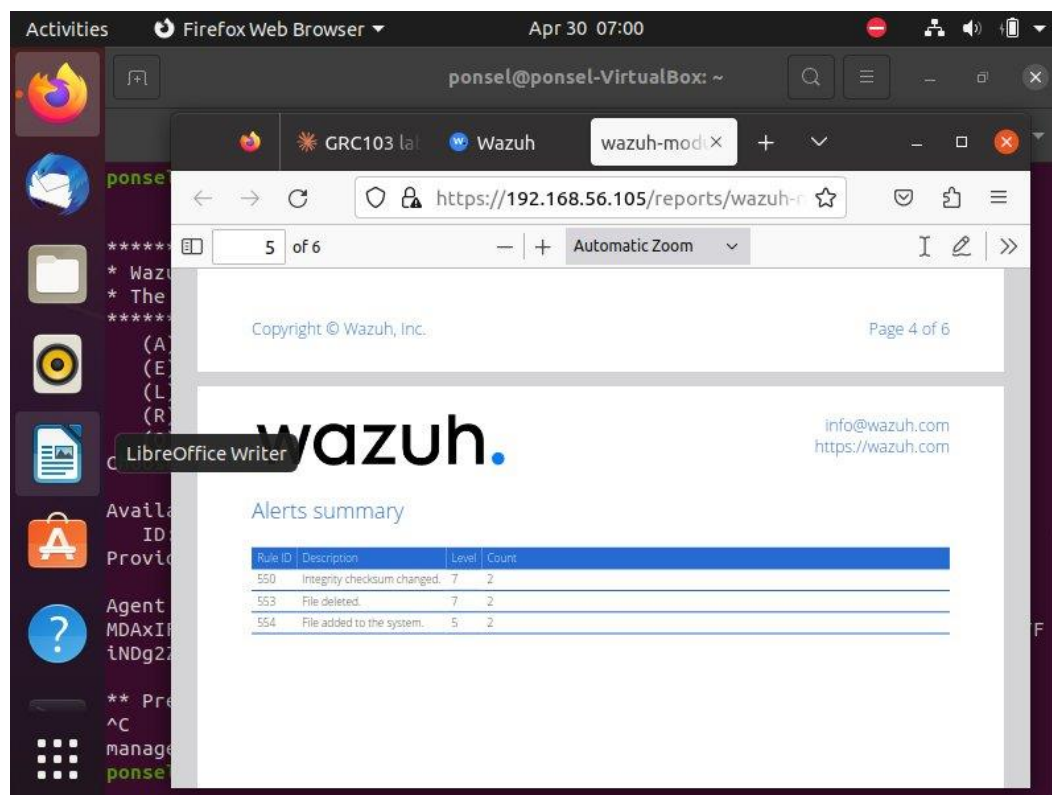


Figure 15: Alert summary showing all 6 detected events with rule descriptions, levels, and counts

6.4 Compliance Mapping

Tools like Wazuh provide direct evidence for audits under frameworks like SOX, PCI DSS, and HIPAA by generating timestamped, immutable logs of every file change on monitored systems. As seen in the Threat Hunting Report, the report shows the agent name, IP address, operating system, and the exact time window of monitoring, which is exactly what an auditor needs.

To demonstrate to an auditor that a specific sensitive file was not altered during the last quarter, you would navigate to the File Integrity Monitoring section, filter by the specific file path such as C:\Users\CONCEPT\SensitiveFiles, and set the date range to the previous quarter. If no modification alerts appear for that file during that period, it serves as evidence that the file remained untouched. This is clearly supported by the checksums and event logs Wazuh records for every monitored file.

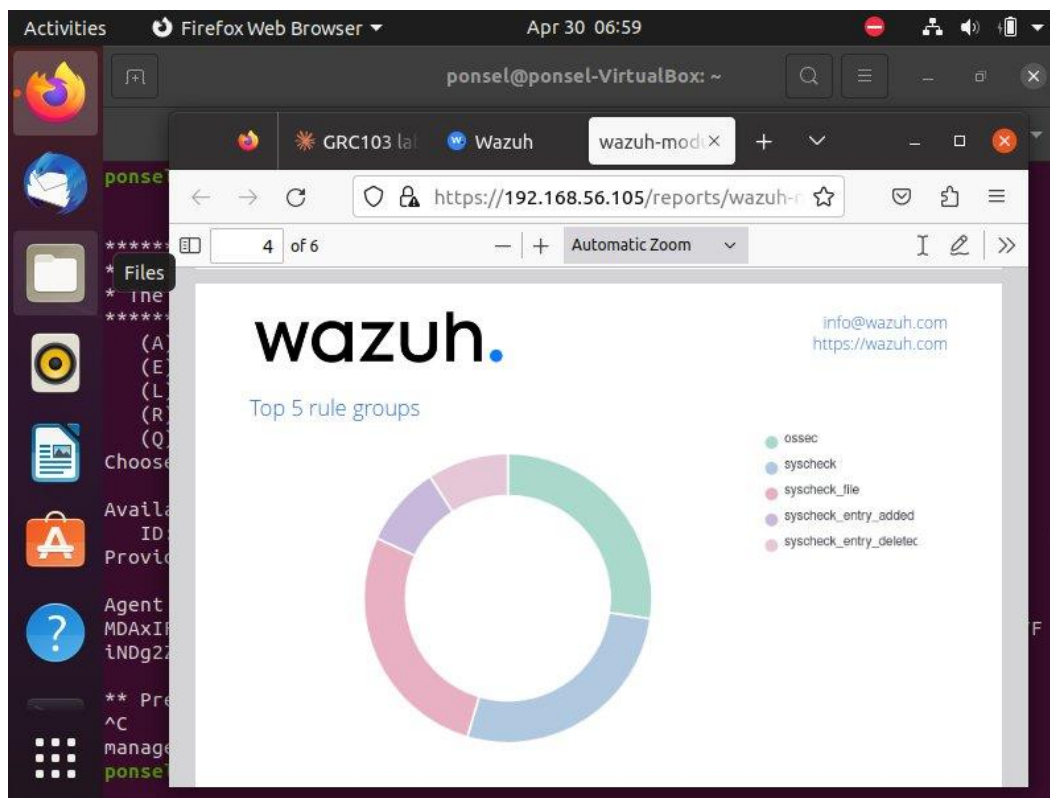


Figure 16: Top 5 Rule Groups chart showing syscheck activity – demonstrating the control's continuous monitoring capability for compliance purposes

7. Conclusion

This lab successfully demonstrated how a technical detective control can be deployed to monitor an IT asset for a specific risk. The Wazuh platform was installed and configured to monitor the SensitiveFiles directory on a Windows workstation in real time. All simulated risk events were detected and logged, generating a complete audit trail with timestamps, rule IDs, and file checksums. The Wazuh dashboard and generated report provide the evidence necessary for compliance reporting and ongoing risk management, bridging the gap between technical monitoring and GRC governance requirements.